

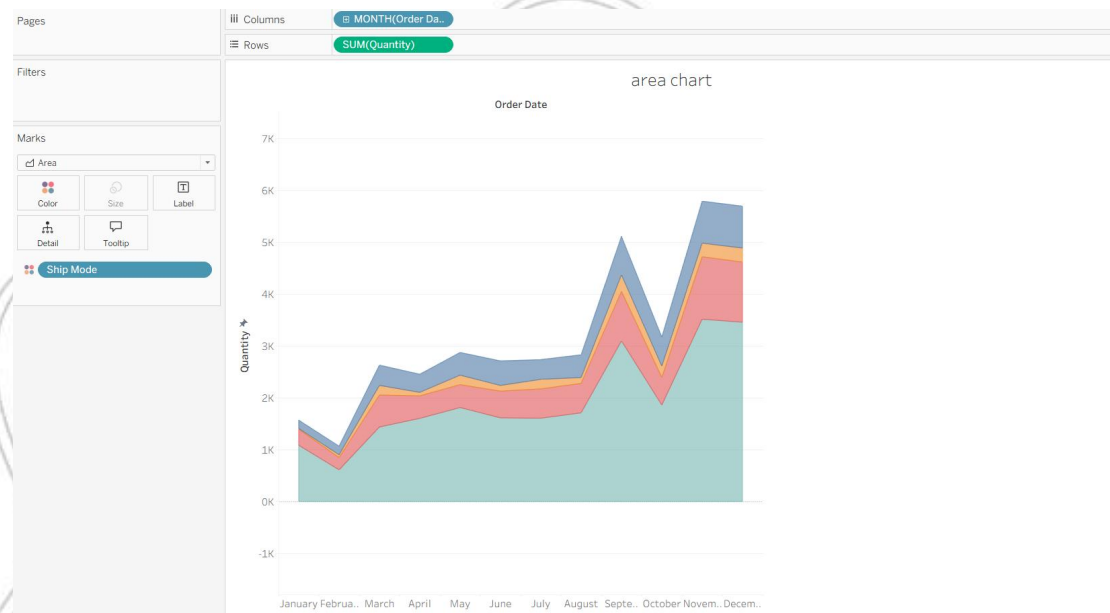
|                    |                       |
|--------------------|-----------------------|
| STUDENT NAME       | KANCHANA<br>YUGANDHAR |
| SECTION            | B.S.c(CS)             |
| HALL TICKET NUMBER | 0325046007            |

## **DATA ANALYTICS WITH TABLEAU ASSIGNMENT**

### **1 – Area Chart**

1. Open Tableau and create a new worksheet from the bottom tab section.
2. In the Data pane, locate Order Date under Dimensions.
3. Drag Order Date onto the Columns shelf.
4. Tableau automatically displays YEAR(Order Date).
5. Click the dropdown arrow on YEAR(Order Date).
6. Choose Month so the data is displayed month-wise instead of year-wise.
7. Locate Quantity under Measures.
8. Drag Quantity onto the Rows shelf.
9. A default line chart may appear automatically.
10. Go to the Marks card and click the chart type dropdown currently set to Automatic.
11. Select Area from the visualization list.
12. Locate Ship Mode under Dimensions.
13. Drag Ship Mode onto Color in the Marks card.
14. Different shipping modes are now displayed with separate colored areas.
15. Resize the worksheet if needed to improve visibility.
16. Hover over the colored areas to view monthly quantity details.

## OUTPUT :

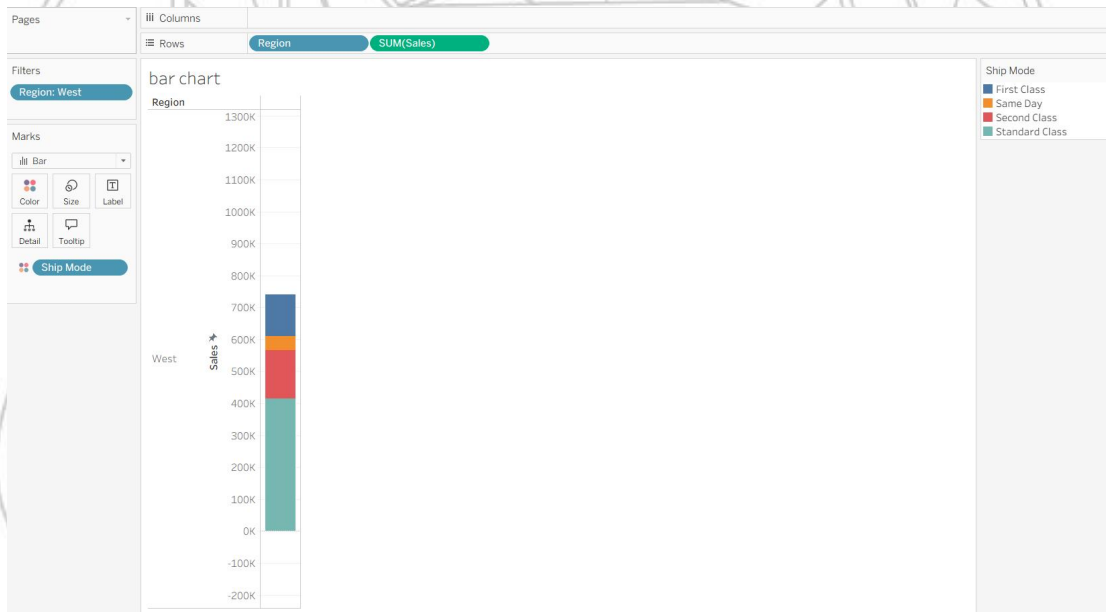


## 2 – Bar Chart

1. Open a new worksheet for the Bar Chart.
2. Locate Sales under Measures.
3. Drag Sales to the Rows shelf.
4. Tableau may initially generate a line chart because of date relationships.
5. Go to the Marks card and open the chart type dropdown.
6. Select Bar.
7. The visualization changes into a bar chart.
8. Locate Ship Mode under Dimensions.
9. Drag Ship Mode onto Color in the Marks card.
10. Each shipping method now appears in a different color.
11. Locate Region under Dimensions.
12. Drag Region to the Rows shelf before SUM(Sales).
13. The chart now compares sales by region.
14. Drag Region again onto the Filters shelf.

15. In the filter dialog box uncheck Central, East, and South.
16. Keep only West selected.
17. Click OK to apply the filter.
18. The chart now displays only the West region sales.

OUTPUT :



### 3 – Box Plot

1. Create a new worksheet for the Box Plot.
2. Drag Segment from Dimensions onto the Columns shelf.
3. Drag Discount from Measures onto the Rows shelf.
4. Right-click SUM(Discount).
5. Go to Measure and choose Average.
6. The aggregation changes from SUM to AVG.
7. Drag Region onto the Columns shelf beside Segment.
8. Open the Show Me panel from the top-right corner.

9. Select Box-and-Whisker Plot.
10. Tableau creates the box plot visualization.
11. Go to the Analysis menu at the top.
12. Uncheck Aggregate Measures.
13. This allows individual data points to be included in the chart.
14. Click the Swap Rows and Columns button to change orientation.
15. Observe the spread, median, and outliers in discount distribution.

**OUTPUT:**

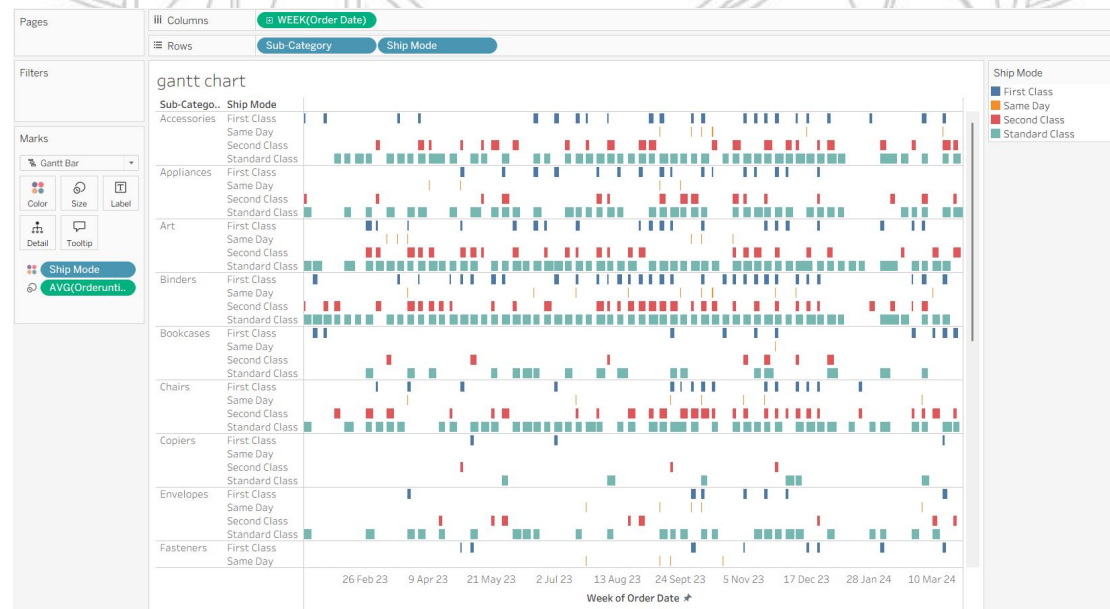


## 4 – Gantt Chart

1. Open a fresh worksheet for the Gantt Chart.
2. Drag Order Date to the Columns shelf.
3. Click the dropdown on YEAR(Order Date).
4. Choose More → Week Number.
5. The timeline now displays weekly intervals.
6. Drag Sub-Category to the Rows shelf.

7. Drag Ship Mode beside Sub-Category on Rows.
8. The rows now display shipping modes within each sub-category.
9. Go to the Marks card.
10. Change the Marks type from Automatic to Gantt Bar.
11. Open the Analysis menu.
12. Select Create Calculated Field.
13. Name the field OrderUntilShip.
14. Enter the formula: DATEDIFF('day',[Order Date],[Ship Date])
15. Click OK.
16. Drag OrderUntilShip onto Size in the Marks card.
17. Bars now represent shipping delay duration.
18. Right-click SUM(OrderUntilShip).
19. Choose Measure → Average.
20. Drag Ship Mode onto Color for visual differentiation.

OUTPUT:



— BUILDERS —

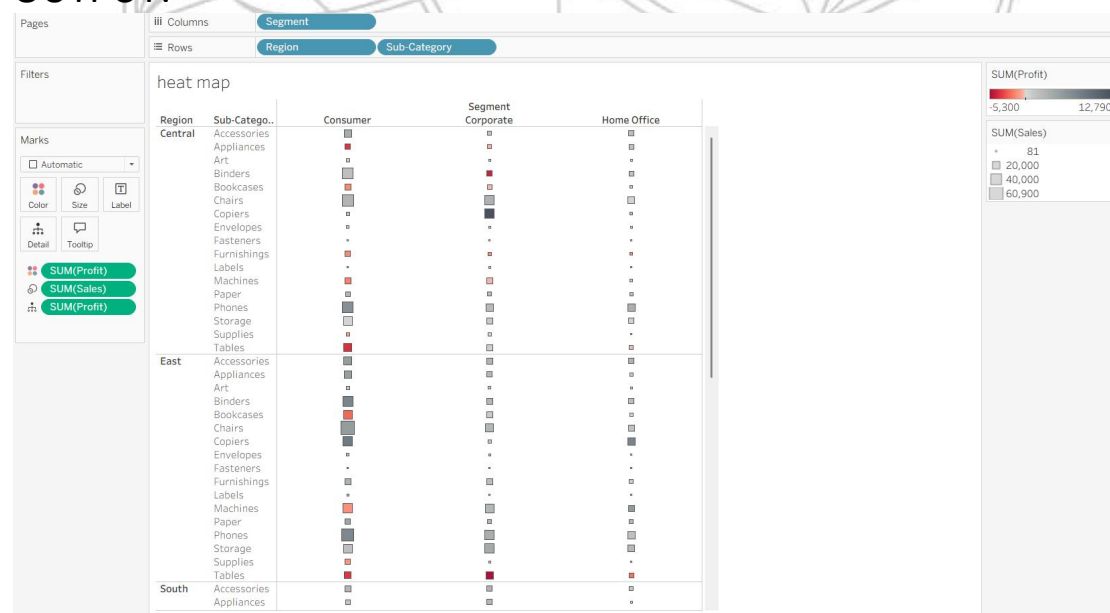




## 5 – Heat Map

1. Create a new worksheet for the Heat Map.
2. Drag Segment onto the Columns shelf.
3. Drag Region onto the Rows shelf.
4. Drag Sub-Category beside Region on Rows.
5. The chart now creates a matrix layout.
6. Drag Profit onto Color in the Marks card.
7. Cells now display color intensity based on profit.
8. Click Color in the Marks card.
9. Choose Border and select Medium Gray.
10. This improves cell visibility.
11. Click the Profit color legend dropdown.
12. Choose Edit Colors.
13. Select a diverging palette such as Red-Black Diverging.
14. Enable Use Full Color Range.
15. Click Apply and OK.
16. Drag Sales onto Size.
17. Larger boxes now represent higher sales values.
18. Click Size and increase the slider for better visibility.

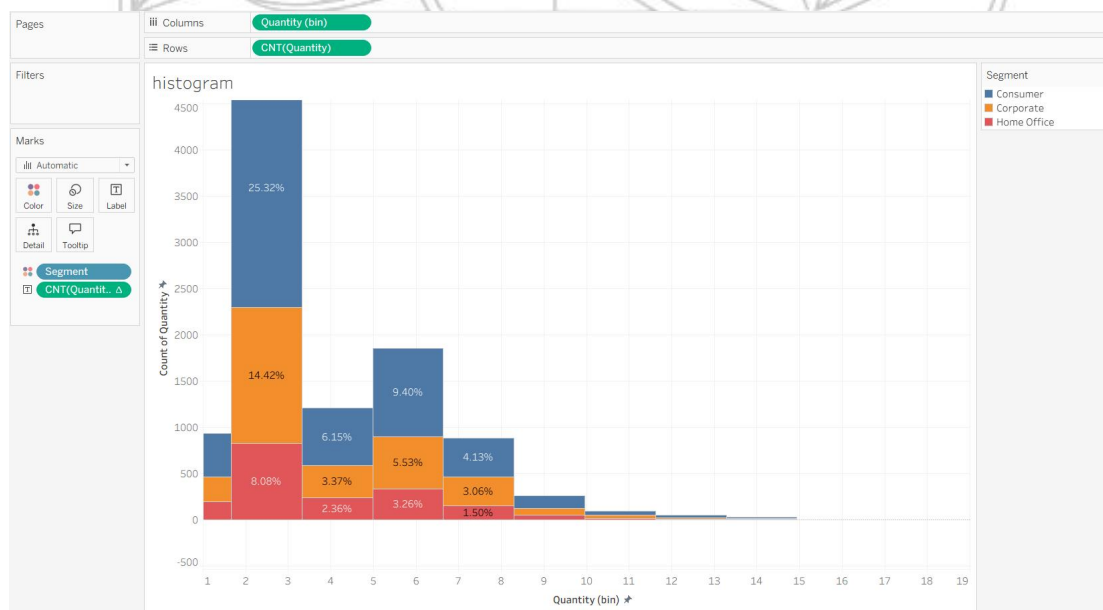
### OUTPUT:



## 6 – Histogram

1. Open a new worksheet for the Histogram.
2. Drag Quantity from Measures onto the Columns shelf.
3. Open the Show Me panel.
4. Select Histogram.
5. Tableau automatically creates Quantity bins.
6. Quantity count is added to Rows automatically.
7. Observe the distribution of quantity values.
8. Drag Segment onto Color in the Marks card.
9. The histogram now separates data by customer segments.
10. Hold Ctrl and drag CNT(Quantity) from Rows to Label.
11. This copies the field without removing it.
12. Right-click CNT(Quantity) in the Marks card.
13. Choose Quick Table Calculation → Percent of Total.
14. Labels now display percentages instead of raw counts.

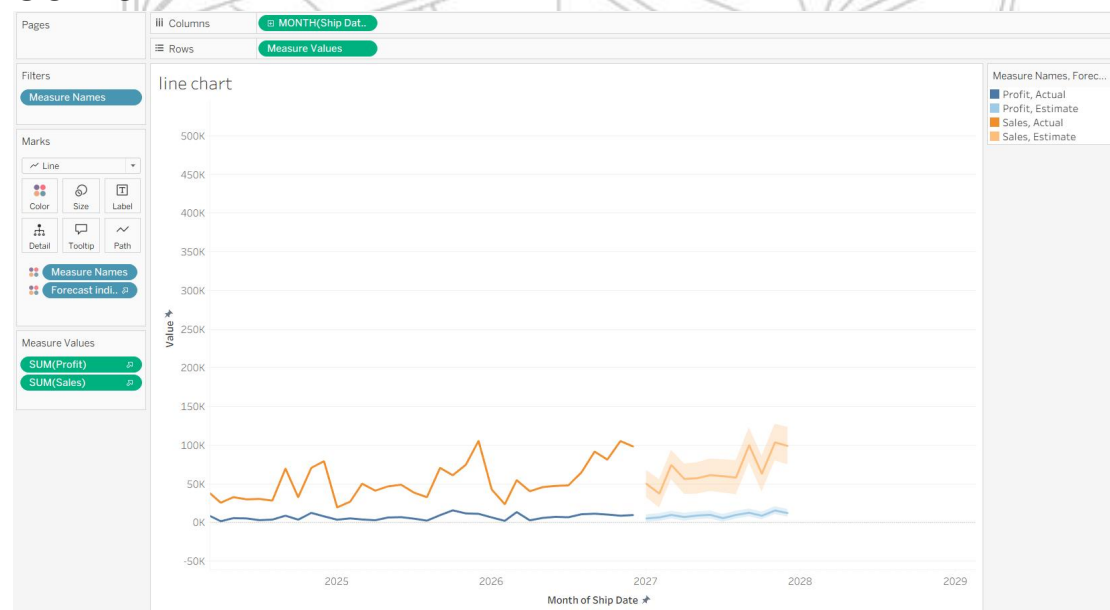
OUTPUT:



## 7 – Line Chart

1. Create a new worksheet for the Line Chart.
2. Drag Ship Date from Dimensions to the Columns shelf.
3. Click the dropdown on YEAR(Ship Date).
4. From the lower green section choose Month.
5. The date field becomes continuous and green.
6. Drag Sales onto the Rows shelf.
7. A line chart displaying sales trends is generated.
8. Drag Profit onto the Rows shelf beside Sales.
9. Two separate axes appear.
10. Drag the Profit axis directly onto the Sales axis until dual-axis indicators appear.
11. Release the mouse to combine both measures.
12. Ensure the Marks type is set to Line.
13. Open the Analytics pane beside Data.
14. Drag Forecast into the chart area.
15. Drop it onto Forecast.
16. Tableau automatically predicts future values based on historical data.
17. Hover over forecasted areas to view projected trends.

### OUTPUT:

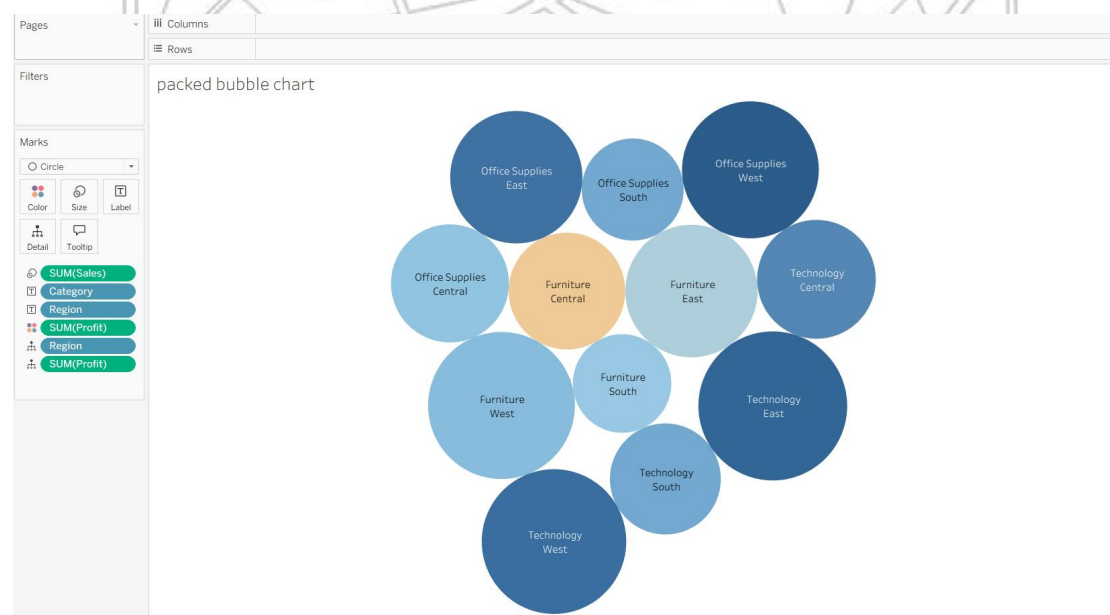




## 8 – Packed Bubble Chart

1. Open a new worksheet for the Packed Bubble Chart.
2. Drag Category onto the Columns shelf.
3. Drag Sales onto the Rows shelf.
4. Open the Show Me panel.
5. Select Packed Bubbles.
6. Tableau converts the chart into packed circles.
7. Bubble size now represents Sales.
8. Drag Region onto Detail in the Marks card.
9. More bubbles appear for regional breakdown.
10. Drag Profit onto Color.
11. Bubble color now represents profitability.
12. Drag Region onto Label.
13. Each bubble now displays the region name.

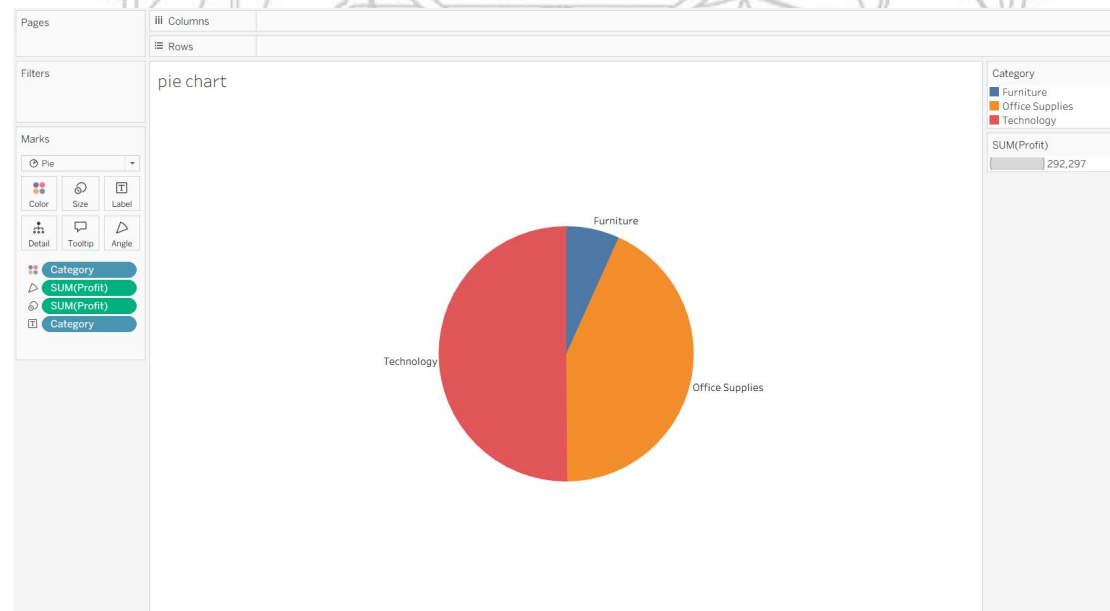
OUTPUT:



## 9 – Pie Chart

1. Create a new worksheet for the Pie Chart.
2. Drag Profit onto the Columns shelf.
3. Drag Category onto the Rows shelf.
4. Open the Show Me panel.
5. Select Pie Chart.
6. The Marks type changes to Pie automatically.
7. Profit controls angle and size by default.
8. Click Fit and choose Entire View.
9. This enlarges the chart area.
10. Click Size in the Marks card.
11. Increase the slider to enlarge the pie chart.
12. Drag Category onto Label.
13. Each slice now displays its category name.

OUTPUT:



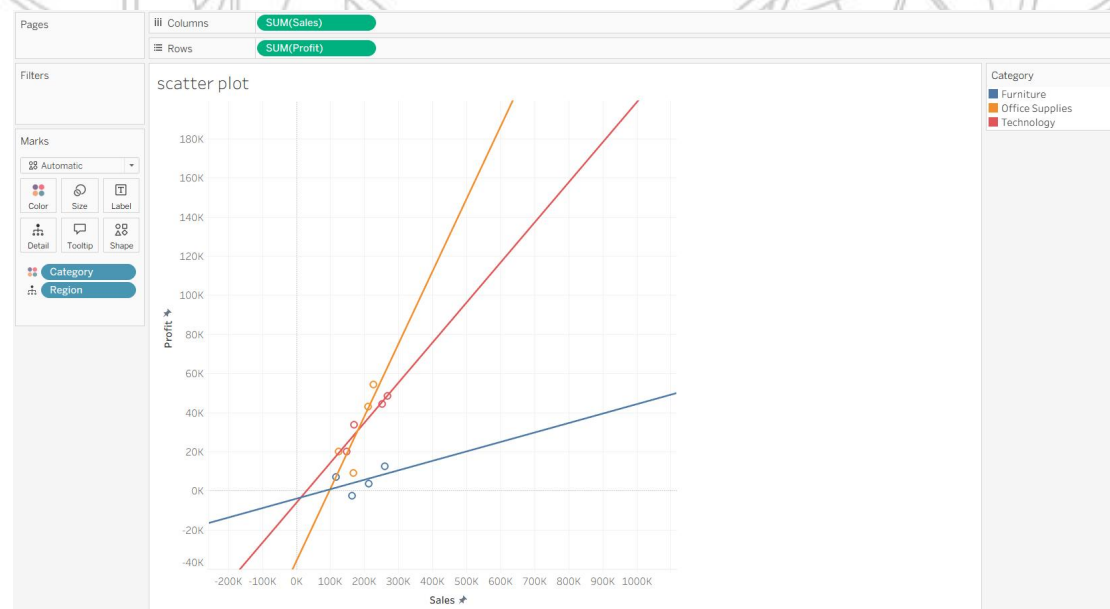
— BUILDERS —



## 10 – Scatter Plot

1. Open a new worksheet for the Scatter Plot.
2. Drag Sales onto the Columns shelf.
3. Drag Profit onto the Rows shelf.
4. A scatter plot framework appears.
5. Drag Category onto Color.
6. Different categories now appear in separate colors.
7. Drag Region onto Detail.
8. Additional marks appear for regional segmentation.
9. Open the Analytics pane.
10. Drag Trend Line onto the chart.
11. Select Linear trend line.
12. Observe the relationship between Sales and Profit.

OUTPUT:



— BUILDERS —

## 11 – Text Table with Percentages

1. Create a new worksheet for the Text Table.
2. Drag Order Date onto the Columns shelf.
3. Tableau displays YEAR(Order Date).
4. Drag Sub-Category onto Rows.
5. A table layout is created.
6. Drag Sales onto Text in the Marks card.
7. Each table cell now displays SUM(Sales).
8. Drag Region before Sub-Category on Rows.
9. The table now groups sub-categories by region.
10. Right-click SUM(Sales).
11. Choose Add Table Calculation.
12. Select Percent of Total as the calculation type.
13. Set Compute Using to Pane Down.
14. Click OK.
15. The values now display percentages instead of raw sales.

OUTPUT:

text table with percentages

|       |             | Order Date |        |        |        |
|-------|-------------|------------|--------|--------|--------|
|       |             | 2023       | 2024   | 2025   | 2026   |
| East  | Storage     | 12.07%     | 8.48%  | 12.11% | 9.49%  |
|       | Supplies    | 0.36%      | 0.76%  | 1.14%  | 3.25%  |
|       | Tables      | 8.36%      | 5.67%  | 4.28%  | 5.48%  |
|       | Accessories | 5.39%      | 5.80%  | 10.02% | 6.64%  |
| South | Appliances  | 2.04%      | 5.40%  | 5.99%  | 6.47%  |
|       | Art         | 0.55%      | 1.91%  | 1.54%  | 1.05%  |
|       | Binders     | 8.00%      | 18.87% | 4.39%  | 9.07%  |
|       | Bookcases   | 0.70%      | 1.74%  | 3.96%  | 4.20%  |
|       | Chairs      | 12.59%     | 14.18% | 6.76%  | 12.74% |
|       | Copiers     | 0.46%      | 1.26%  | 2.73%  | 4.36%  |
|       | Envelopes   | 0.24%      | 1.97%  | 0.96%  | 0.64%  |
|       | Fasteners   | 0.16%      | 0.18%  | 0.11%  | 0.09%  |
|       | Furnishings | 3.04%      | 7.54%  | 5.75%  | 2.75%  |
|       | Labels      | 0.39%      | 1.57%  | 0.31%  | 0.44%  |
|       | Machines    | 26.57%     | 4.36%  | 14.14% | 8.09%  |
|       | Paper       | 3.07%      | 3.23%  | 4.19%  | 3.86%  |
| West  | Phones      | 16.61%     | 11.00% | 12.65% | 17.38% |
|       | Storage     | 6.47%      | 10.49% | 11.58% | 8.73%  |
|       | Supplies    | 4.09%      | 0.18%  | 1.56%  | 2.02%  |
|       | Tables      | 9.57%      | 10.33% | 13.35% | 11.48% |
|       | Accessories | 5.91%      | 7.53%  | 8.19%  | 10.08% |
|       | Appliances  | 2.50%      | 5.45%  | 2.65%  | 5.35%  |
|       | Art         | 2.25%      | 1.48%  | 0.60%  | 1.04%  |
|       | Binders     | 8.59%      | 2.74%  | 6.65%  | 10.28% |
|       | Bookcases   | 4.34%      | 6.60%  | 4.39%  | 4.70%  |
|       | Chairs      | 15.42%     | 17.37% | 17.01% | 10.49% |
|       | Copiers     | 1.99%      | 5.30%  | 5.18%  | 11.41% |
|       | Envelopes   | 0.44%      | 0.98%  | 0.83%  | 0.21%  |
|       | Fasteners   | 0.19%      | 0.10%  | 0.14%  | 2.11%  |
|       | Furnishings | 3.25%      | 6.07%  | 4.34%  | 4.06%  |
|       | Labels      | 0.58%      | 0.72%  | 0.66%  | 0.75%  |
|       | Machines    | 2.55%      | 5.91%  | 7.83%  | 6.24%  |
|       | Paper       | 3.57%      | 3.88%  | 3.43%  | 3.62%  |
|       | Phones      | 19.59%     | 13.09% | 13.72% | 9.58%  |

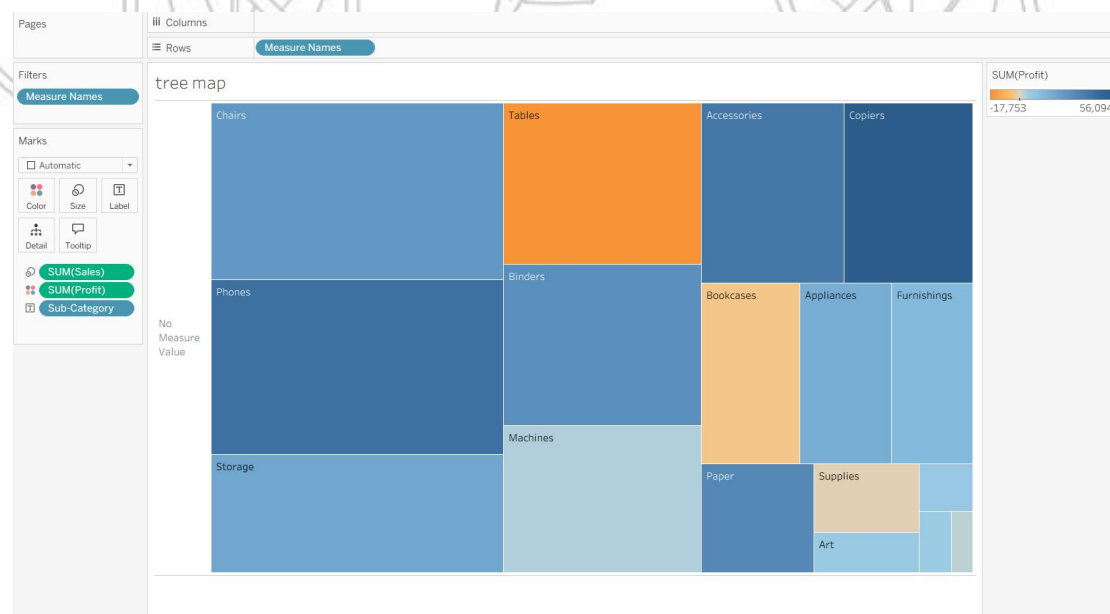
— BUILDERS —



## 12 – Treemap

1. Open a new worksheet for the Treemap.
2. Drag Sub-Category onto the Columns shelf.
3. Drag Sales onto the Rows shelf.
4. Open the Show Me panel.
5. Select Treemap.
6. Rectangles now represent sub-categories.
7. Rectangle size is based on Sales.
8. Drag Ship Mode onto Color.
9. Rectangles now display different colors for shipping modes.
10. Optionally drag Profit onto Color instead of Ship Mode.
11. This changes color representation from shipping mode to profitability.

OUTPUT:



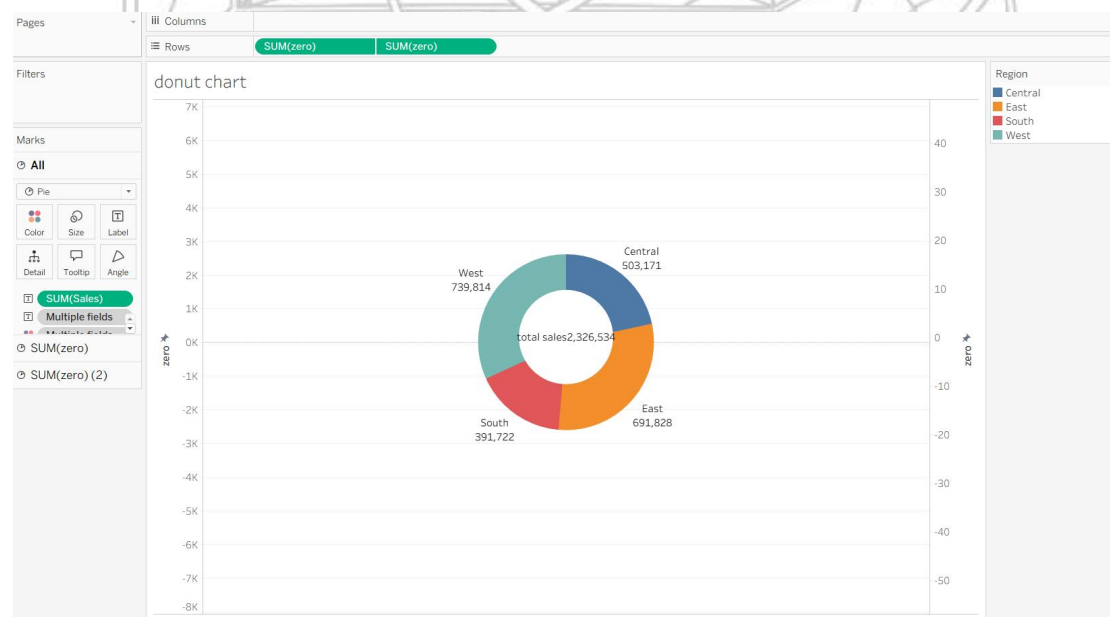
## 13 – Donut Chart

1. Create a new worksheet for the Donut Chart.
2. Go to Analysis → Create Calculated Field.
3. Name the field Zero.



4. Enter the value 0 and click OK.
5. Drag Zero onto the Rows shelf.
6. Change the Marks type to Pie.
7. Click Fit → Entire View.
8. Increase the pie size using the Size slider.
9. Drag Zero again onto the Rows shelf beside the first Zero field.
10. Right-click the second SUM(Zero).
11. Select Dual Axis.
12. Two overlapping pie charts now appear.
13. Select the second Marks card.
14. Reduce the size slightly.
15. Change its color to White.
16. This creates the donut hole effect.
17. Select the first Marks card.
18. Drag Region onto Color.
19. Drag Sales onto Angle and Label.
20. Add Total Sales text to the center using Label formatting.

## OUTPUT:



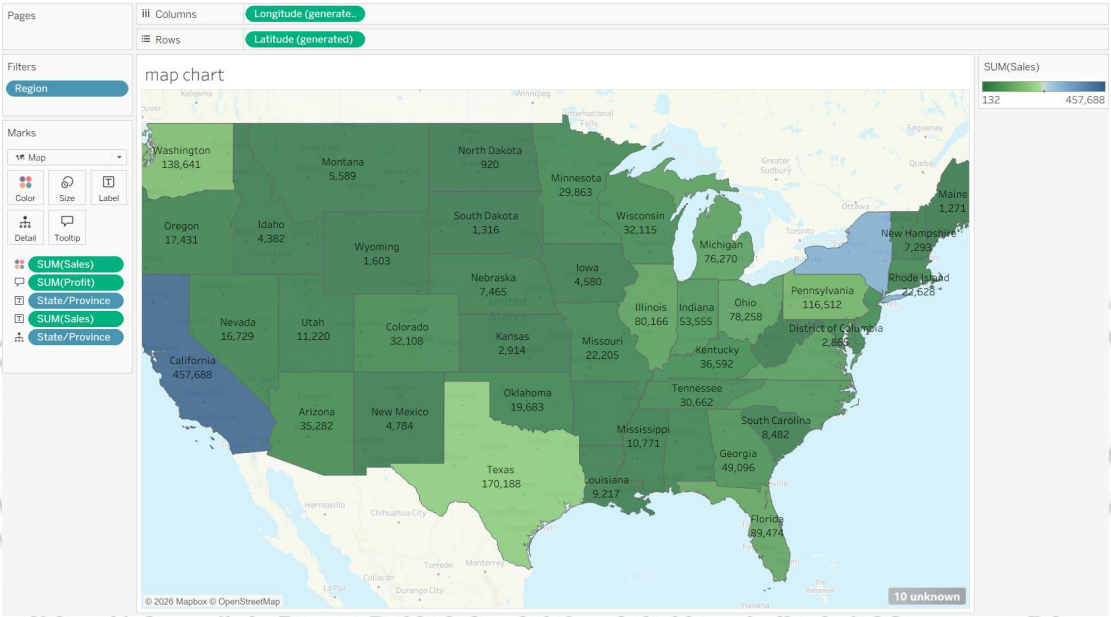
## 14 – Map Chart

1. Open a new worksheet for the Map Chart.
2. Locate State/Province under Dimensions.
3. Right-click State/Province.
4. Verify the Geographic Role is set to State/Province.
5. Double-click State/Province.
6. Tableau automatically generates a map using Latitude and Longitude.
7. If only a blank world map appears, change the Marks type to Filled Map.
8. Drag Sales onto Color.
9. States now display colors based on sales values.
10. Click Color → Edit Colors.
11. Choose a suitable palette such as Purple or Blue.
12. Click OK.
13. Drag State/Province onto Label.
14. Drag Sales onto Label.
15. Each state now displays its sales amount.
16. Drag Region onto the Filters shelf.
17. Select all regions and click OK.
18. Right-click Region and choose Show Filter.
19. Users can now interactively filter the map.
20. Drag Profit onto Tooltip.
21. Hovering over states now displays profit details.
22. Go to Map → Map Layers.
23. Set Background Style to Outdoors.
24. Adjust Washout to approximately 70% for improved readability.

BRAVO  
— BUILDERS —



OUTPUT:



DASHBOARDS

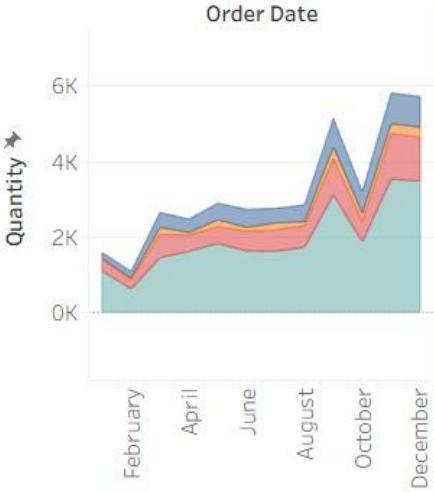
BRAVO

— BUILDERS —

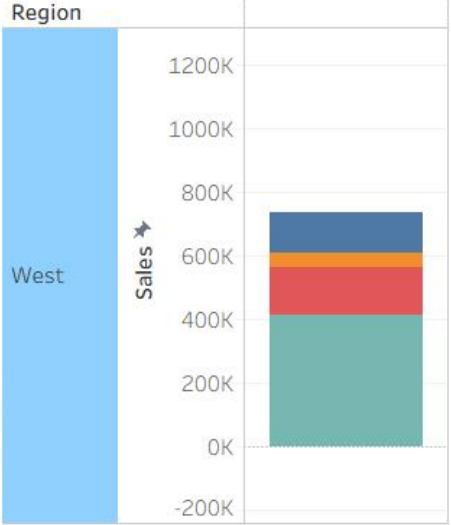


# DASHBOARD 01

area chart



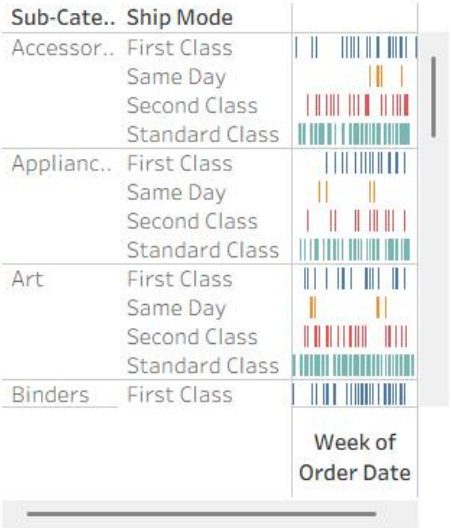
bar chart



box plot



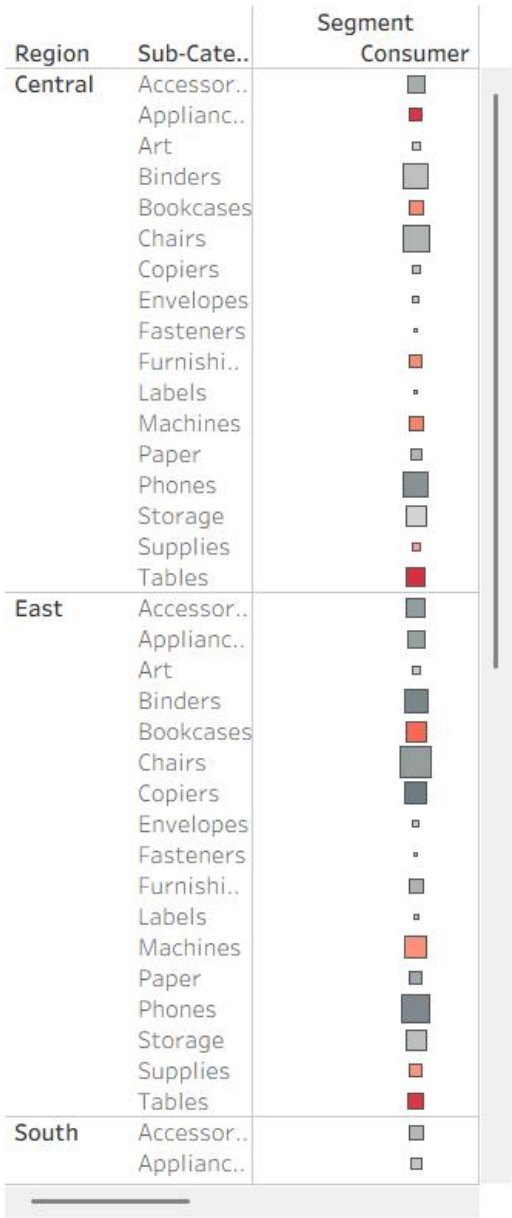
gantt chart



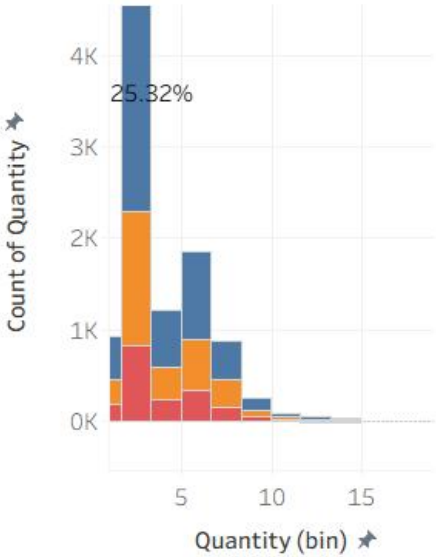


DASHBOARD 02

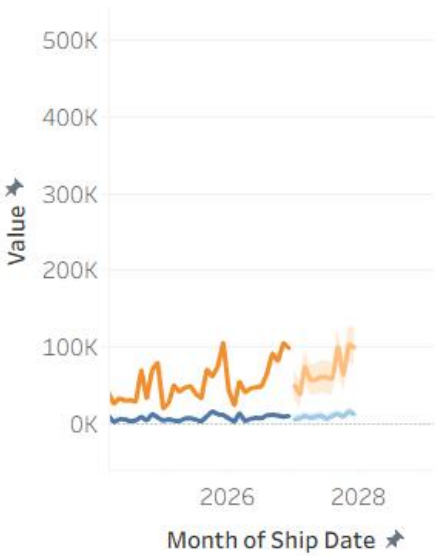
heat map



histogram



line chart

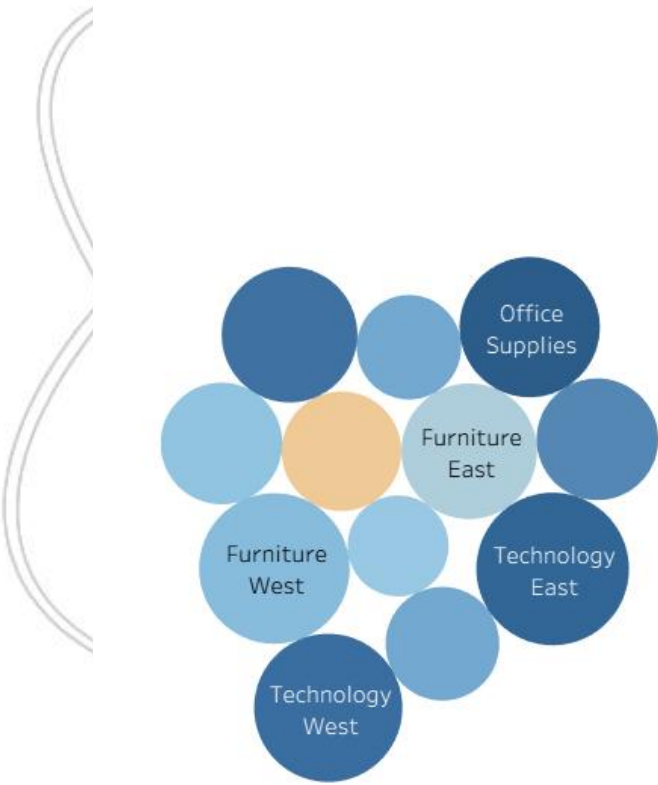


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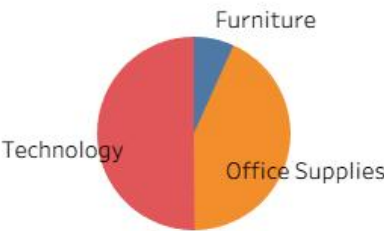


DASHBOARD 03

packed bubble chart



pie chart



scatter plot



— BUILDERS —

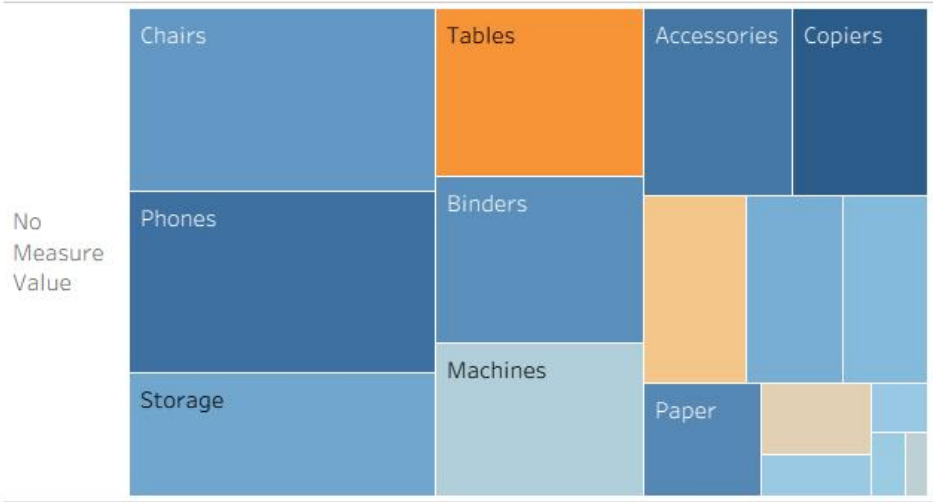


DASHBOARD 04

text table with percentages

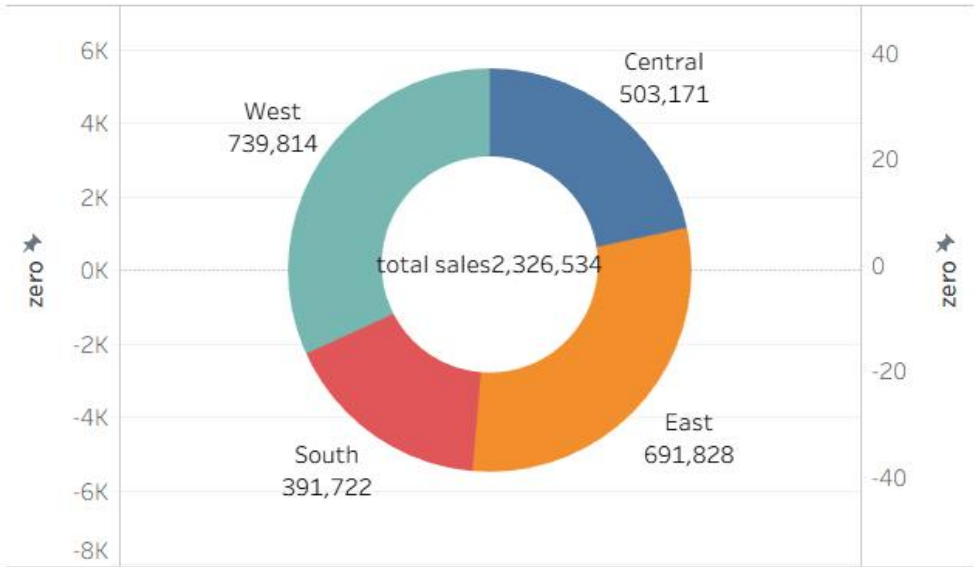
| Region  | Sub-Cate.. | Order Date |      |
|---------|------------|------------|------|
|         |            | 2023       | 2024 |
| Central | Accessor.. | 4.27%      | 7.   |
|         | Applianc.. | 3.52%      | 4.   |
|         | Art        | 0.79%      | 1.   |
|         | Binders    | 15.28%     | 5.   |
|         | Bookcases  | 1.77%      | 8.   |
|         | Chairs     | 19.99%     | 17.  |
|         | Copiers    | 3.15%      | 12.  |
|         | Envelopes  | 1.54%      | 0.   |
|         | Fasteners  | 0.12%      | 0.   |
|         | Furnishi.. | 2.44%      | 2.   |
|         | Labels     | 1.01%      | 0.   |
|         | Machines   | 15.69%     | 1.   |
|         | Paper      | 2.26%      | 3.   |
|         | Phones     | 9.56%      | 18.  |
|         | St...      | 10.60%     | 0.   |

tree map

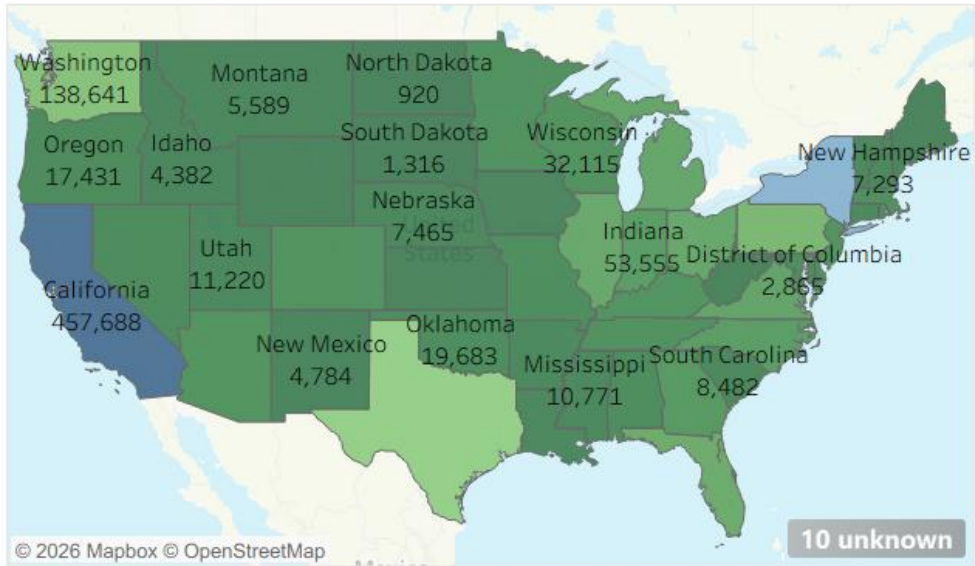


# DASHBOARD 05

donut chart

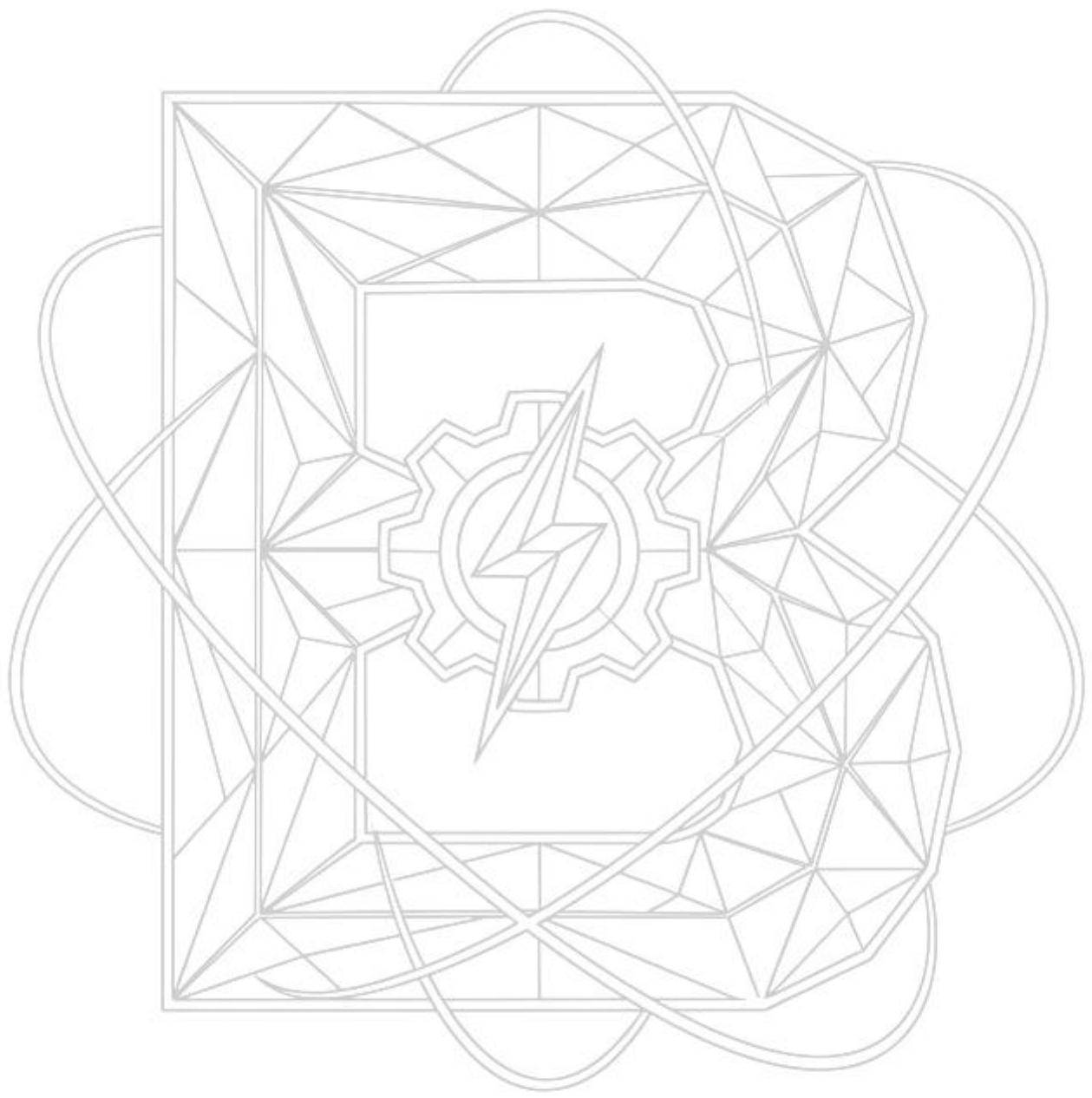


map chart



— BUILDERS —





BRAVO  
— BUILDERS —

